Identifiability of independent sampling units in publicly deposited metabolomics data: a repository-level audit and documentary assessment

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Abstract

Introduction. Reuse of public metabolomics data requires that independent sampling units be identifiable; unique sample labels do not identify distinct participants. **Objectives.** To assess whether repository metadata and linked documentation establish participant-level sampling structure under prespecified, affirmative evidence rules. **Methods.** A metadata-only census of 3,582 public MetaboLights accessions; a secondary audit of factor metadata in 104 previously selected Metabolomics Workbench studies; a randomised documentary assessment of 40 of 371 unresolved MetaboLights candidates, reading deposit metadata with linked publications; and a pre-verification of 19 human deposits removed by an automated quality-control rule. Source inventories were frozen and signed before reading, grading was deterministic, extraction was AI-assisted, and the author reviewed a prespecified subset. **Results.** Of 797 human single-matrix MS or NMR studies reaching the census independence test, none met the criteria. No randomly selected study met the strict one-sample-per-participant standard (0 of 40; one-sided 95% upper bound 25 of 371 candidates). Documentation described repeated sampling in 9 of 40 studies (8 on re-review) and cell lines in 3. In 17 of 19 quality-control-excluded deposits, other fields identified study samples rather than blanks. The author's review disagreed with the AI extraction only on assigning counts to the selected assay, never on the strict result. **Conclusion.** Under prespecified documentary criteria, participant-level sampling independence was not established in the census or the randomised assessment. The findings concern what documentation establishes, not true sample independence. Deposition standards should require explicit participant-to-sample linkage.

Keywords: metabolomics, data reuse, external validation, metadata, repositories, pseudoreplication

1 Introduction

Public repositories have made thousands of metabolomics studies available for reuse. MetaboLights and the Metabolomics Workbench together hold several thousand studies, and their deposits are increasingly reused for method development, benchmarking and the evaluation of predictive models (Haug et al, 2020; Sud et al, 2016). A common form of reuse treats each deposited study as an independent data set and each deposited sample as an observation. That treatment is valid only if an analyst can tell which samples come from the same participant.

The requirement is not that every study contribute one sample per participant. Longitudinal and repeated-measures designs are legitimate and often informative, and they can support external validation when the dependence among a participant's samples is modelled, clustered or reduced to one prespecified observation per participant. What any valid reuse needs is that the independent observational units be identifiable: the deposit, or the documentation linked to it, must allow the analyst to determine how many participants contributed, which samples belong to which participant, and whether any participant contributed more than one sample to the analysed set. When this information is missing, a repeated-measures study is indistinguishable from a cross-sectional one, and analyses that treat rows as independent participants risk pseudoreplication (Hurlbert, 1984; Lazic, 2010). In multi-study external validation (Riley et al, 2016) the same problem arises one level up: studies that share participants are not independent validation units.

The consequences of undetected dependence are documented in other omics fields. In transcriptomics, hidden duplicate samples across public data sets are common, and in simulation they inflate apparent cross-study validation performance (Waldron et al, 2016). Sample-level misannotation is also widespread in public expression data (Toker et al, 2016). Controlled-access genomic repositories address this structurally: dbGaP, for example, requires a subject–sample mapping file for molecular data and allows one subject to map to several samples (National Center for Biotechnology Information, n.d.).

For metabolomics repositories, metadata quality has been assessed in general terms. Compliance with the minimum reporting standards of the Metabolomics Standards Initiative is incomplete and uneven across repositories (Spicer et al, 2017a,b), automated quality scores for MetaboLights metadata vary widely (Ferreira et al, 2017), sample identifiers in Metabolomics Workbench files do not always match across sections (Powell and Moseley, 2021), and only a minority of public mass-spectrometry files meet controlled-vocabulary metadata templates (Jarmusch et al, 2020). These studies show that metadata are often incomplete. We are not aware of one that measures, under a prespecified evidence rule, the specific property that study-level reuse depends on: whether the deposited information identifies the independent sampling units.

Here we measure that property directly. We define, before examining any study, what counts as affirmative evidence that the participants and samples of a deposited assay can be identified, and we apply the definition in four complementary ways, each with its own denominator and selection procedure: (i) a census of all 3,582 public MetaboLights accessions available on the census date, using deposited metadata only;

(ii) an audit of subject-level fields in the factor metadata of 104 human Metabolomics Workbench studies; (iii) a documentary assessment of a randomly selected tranche of 40 MetaboLights candidates, reading the deposit metadata together with the methods sections of the linked publications; and (iv) a pre-verification of 19 human MetaboLights deposits whose sample rows an automated quality-control rule had removed in full. Throughout, we distinguish evidence that independence is established by the available documentation from the true independence of the underlying samples: failure to establish the former is not evidence against the latter.

2 Methods

All four analyses were metadata- and documentation-only. No measured metabolite value, data matrix, raw or derived data file was opened at any stage. Each analysis was specified in a written protocol, implemented in code with an offline test suite, and bound by SHA-256 hashes in a lock file signed by the author before execution. The locks were not registered with an external registry. Protocols, code, locks and outputs are public (Declarations).

2.1 Target property and evidence standard

The unit of interest is a deposited assay: the biological samples measured on one platform for one matrix. We ask whether the deposited metadata or linked documentation identify its independent sampling units: whether the samples come from human participants, how many distinct participants contributed, and whether any contributed more than one sample (time points, pre/post samples, several specimens, replicates entered as biological samples) or, failing that, whether a sample-to-participant mapping lets an analyst model the dependence. Evidence had to be affirmative: unique sample or source labels were bookkeeping, not evidence; counts had to be stated integers or sums over groups the source called disjoint; an enrolment figure counted only if the text tied it to the analysed samples. Independence established in this sense is "documented"; a study whose documentation is silent may still consist of independent participants.

2.2 MetaboLights census

A locked enumeration audit fixed the universe at 3,582 accessions (REST list $\cap$ FTP mirror, minus two development exemplars) on 25 October 2026. The census read only investigation, sample and assay files; data files were refused by an allow-list guard. Studies were screened in a fixed order (human organism; removal of quality-control, blank, pooled and reference rows; sample-name integrity; a single matrix; the independence test). The independence test had two affirmative routes: a person-identifier column with exactly one sample per identifier (R1), or a cross-sectional design annotation plus a study-description sentence matching one of four frozen one-sample-per-participant templates, guarded against negation (R2); both required at least 50 sources. The strict criterion (L1) counted R1 and explicit R2 statements; a sensitivity count added count-only statements. Studies that failed the test but had

at least 50 distinct source labels with one sample each, or at least 50 samples with a degenerate source column, became unresolved candidates for documentary assessment (Supplementary Methods S1).

2.3 Secondary audit of Metabolomics Workbench subject fields

This analysis is secondary. It reuses 104 human Metabolomics Workbench studies retained on 17 October 2026 by an earlier value-blind screen built for a different, subsequently closed research question; its criteria and exclusion counts, copied from its hashed closure record, are in Supplementary Table S1. The 104 are not a representative sample of the Workbench, and no inference is drawn about the repository as a whole. For each study we read the `factors` table returned by the Workbench REST interface (from the screen's cache; no network request) and recorded whether any factor key named a subject, patient, participant, individual, donor, person, volunteer or proband. Workbench mwTab files, which may hold an optional subject column, embed measured values and were not opened, so nothing is claimed about subject fields elsewhere in the Workbench.

2.4 Documentary assessment of a random tranche (T1)

Population and selection.

The census produced 371 unresolved MetaboLights candidates with at least 50 unique source labels (312) or at least 50 samples with a degenerate source column (59). A tranche of 40 was selected as follows. Each candidate was ranked by SHA-256 of a salt concatenated with its accession, and the first 40 were taken. The salt was the output value of a NIST Randomness Beacon (version 2) pulse, at a time fixed in the signed lock before the pulse existed (pulse 2-2005958, 30 October 2026, 15:00 UTC). No candidate was replaced. The tranche is therefore a simple random sample of 40 of the 371 candidates, and finite-population inference applies.

Sources, reading views and document freeze.

Four sources were admissible: the deposit's investigation file; publications named in the deposit, retrieved as full text from Europe PMC; the first five publications returned by a frozen full-text search for the exact accession string, usable only if their text contained it; and author-supplied copies of named publications that were not open access, registered before the freeze. No author was contacted. Extractors read only a deterministic reading view of each publication (title, abstract with results-type parts removed, methods-type sections and participant-characteristics tables). After retrieval, every document and view was listed with its hash in an evidence inventory that the author signed before any evidence was read; nothing could be added afterwards (Supplementary Methods S3).

Questions and grades.

For the census-chosen assay of each study, extractors answered: Q1, whether the samples come from human participants; Q2, whether a source identifies the sample set analysed in the deposited assay; Q3, the number of distinct participants

and whether it is scoped to the analysed set, to enrolment only, or absent; Q4, whether the source states one sample per participant, states repeated sampling, or is silent; Q6, overlap flags with other deposits; and Q7, whether a sample-to-participant mapping is published. Every answer other than "not stated" required a verbatim quotation with its location. A deterministic function assigned grades in a fixed order (Supplementary Table S2). The STRICT grade, which requires a documented one-sample-per-participant structure, was designed for a downstream analysis that needs one sample per participant. It is more restrictive than the question of this paper: a study whose documentation permits an analyst to reconstruct the clustering of repeated samples would not reach STRICT. We therefore also report Q7 (a published sample-to-participant mapping) and the Q4 category, and we do not equate a zero STRICT count with the absence of reconstructible clustering.

Extraction, grading and adjudication.

Extraction was performed by AI sub-agents working under a written brief fixed and hashed before any record was produced (Section 2.8). The extractors were not told the deposits' biological row counts; the correspondence between a stated count and the deposited rows (rules 6–7) was computed only by the grading function. Every quotation was checked automatically for verbatim presence in the cited view. The author adjudicated every CONTRADICTED and STRICT record against the filed evidence; a rejected CONTRADICTED record became UNRESOLVED. No record was edited after grading.

2.5 Pre-verification of deposits removed by quality-control filtering

A diagnostic after the census identified 19 MetaboLights deposits with human organism values and at least 50 sample rows in which every row had been removed by the M2 quality-control rule. Because the census selected no assay for them, the four checks were applied to the biological samples listed in each deposit's sample file: V1, human participants; V2, the rows are identified as participants' samples by fields other than the sample-type label, or by a publication; V3, an integer number of biological samples stated in text (counting rows was not accepted); and V4, the sampling structure, failing on explicit repeated sampling. A study was verified if all four checks passed and failed if any check failed. Sources were the investigation, sample and assay files and named publications. The evidence set was frozen and signed before reading, extraction was blind to row counts, and quotations of sample names or individual-level rows were prohibited.

2.6 Statistical analysis

For the random tranche, the number of STRICT studies among $n = 40$ drawn without replacement from $N = 371$ bounds the number K of documentation-recoverable candidates: the one-sided 95% upper bound is the largest K with $P(X \leq x \mid N, K, n) > 0.05$ under the hypergeometric distribution (Clopper–Pearson bounds scaled by N are given

for comparison). These bound candidates whose documentation meets the standard, not truly independent cohorts. All other results are counts with their denominators.

2.7 Author review of the AI-assisted extraction

A two-part review was specified and hashed on 31 October 2026, before the reviewed studies were identified. The protocol intended an independent human reviewer; none could be recruited, and the author performed both components on 1–2 November 2026. In component A the author re-extracted Q3 and Q4, using the frozen categories, for ten studies selected by a prespecified hash ranking from the 17 tranche studies with publication text that the author had not adjudicated. The author had no access to the original extractor's answers or quotations, but had prior knowledge of the study classifications and had seen the answers of a second AI model run on the same ten studies in an earlier, abandoned attempt at this component (preserved and labelled as an AI extraction); the answer sheet was assembled by script, so per-study completion times are unavailable. Agreement is reported separately for Q3 scope, Q3 count and Q4, with every disagreement listed, and is not extrapolated to the tranche. In component B the author re-read the sources of all 12 CONTRADICTED records and recorded whether an explicit statement supports each classification; it is a non-blind consistency check by the reader who had adjudicated them. Neither component alters the frozen grades; all quotations were machine-checked. Details are in Supplementary Methods S5.

2.8 Use of AI tools

Evidence extraction for the tranche and the pre-verification was performed by sub-agents of Claude (Anthropic), working under fixed, hashed written briefs. Each sub-agent read only the sealed files of its assigned studies, produced structured records with verbatim quotations, and had to pass an automated validator that checked the schema, the verbatim presence of every quotation and the form of every count. Grades were assigned by deterministic code, not by the sub-agents. The author adjudicated every CONTRADICTED record and carried out the human validation in Section 2.7. The briefs, validators and raw sub-agent outputs are public. Claude also assisted with code and drafting under the author's direction; the author is responsible for the content.

3 Results

3.1 MetaboLights census

Of the 3,582 accessions, 3,569 reached a terminal status; 11 failed in transport and 2 returned persistent HTTP 403 responses (Table 1). Most exclusions were non-human studies (2,321). Of the human studies, 797 with a single sample matrix and an MS (714) or NMR (83) assay reached the independence test. **No study met the prespecified affirmative criteria under either the strict or the sensitivity count (L1 = 0).**

Neither route found qualifying evidence. Among the inspected records (the 658 studies whose source-node columns were tested against the R1 pattern and the 220 studies rescreened after all their rows had been classified as quality control), no qualifying person-identifier column was found. The R2 design layer passed in 18 of the

797 studies, and none of the 18 contained a qualifying one-sample-per-participant sentence. The census found 312 human studies with at least 50 distinct source labels and one sample per source, and 59 with at least 50 samples and a degenerate source column; these 371 formed the population for the documentary assessment.

In 220 studies every sample-sheet row was classified as quality control. In 183 of them the trigger was the value `blank` in a sample-type characteristic on every row. The census read the label as written; whether the rows are in fact biological samples was examined for the human subset in Section 3.5.

Table 1 Outcome of the MetaboLights census (3,582 accessions)

Status	Rule or reason	n
Excluded	non-human (M1)	2,321
Excluded	more than one matrix (M4)	67
Excluded	> 1 sample per source or identifier (M5)	60
Excluded	no MS or NMR assay	1
Unresolved	source-unique, failed M5 (312 with ≥ 50 sources)	616
Unresolved	degenerate source column (59 with ≥ 50 samples)	121
Unresolved	M2: all rows QC / ambiguous QC label / unidentified rows	220 / 66 / 7
Unresolved	duplicate sample names / organism / organism part	13 / 3 / 1
Refused	format / structure / index	61 / 10 / 2
Not terminal	transport failure / HTTP 403	11 / 2
Eligible	strict and sensitivity L1	**0**

3.2 Secondary audit of Metabolomics Workbench subject fields

None of the 104 previously selected human studies had a factor key naming a subject, patient, participant, individual, donor, person, volunteer or proband in the `factors` table. The most frequent keys were design factors: sample source (21), treatment (16), group (13), sex or gender (13), diagnosis (8). Two limitations bear on this result. The inherited repeated-measures detector did not recognise underscore or camel-case time keys (for example `POST_SURGERY_TIME`, `collectionWeek`), so dependence in these studies is undercounted. Factor keys such as cell line, cell culture medium and post-mortem interval show that a "human" species label did not by itself establish human participants. These findings concern the `factors` endpoint of 104 selected studies only.

3.3 Documentary assessment of the random tranche

Of the 40 studies, 29 had at least one publication reading view in the frozen evidence set and 11 had none. Under the prespecified procedure, no qualifying statement that each participant contributed one sample to the analysed set was identified in any reading view, and no view contained a sample-to-participant table; Q7 was "none" for all 40 studies. **No study reached the STRICT grade ($x = 0$).** Because STRICT requires a stated one-sample-per-participant structure, this count does not measure how many studies' clustering could be reconstructed from other documentation; that question is

addressed by Q7 and the Q4 categories reported below. The one-sided 95% hypergeometric upper bound on the number of the 371 candidates whose documentation meets the standard is 25 (Clopper–Pearson: 26.8).

Table 2 Final grades of the 40 tranche studies after adjudication

Grade (reason)	n
STRICT	0
COUNT (sensitivity only)	5
CONTRADICTED: repeated sampling in the chosen assay	9
CONTRADICTED: samples not from human participants (cell lines)	3
UNRESOLVED: adjudication rejected	3
UNRESOLVED: assay membership not established	6
UNRESOLVED: no assay-scoped participant count	5
UNRESOLVED: stated count below / above deposited rows	3 / 3
UNRESOLVED: count supported only by the deposit	3

Where documentation addressed sampling, it often described dependence. Nine studies (22.5%) documented repeated sampling of the same participants within the chosen assay (serial time points, pre/post-treatment samples, follow-up visits, several cysts per patient, up to 56 time points per participant), and three (7.5%) consisted of cultured cell lines rather than participant samples. Together these account for the 12 CONTRADICTED classifications after adjudication. All twelve had entered the candidate pool through unique or degenerate source labels. Under the author's re-review (Section 3.4) one of the nine repeated-sampling classifications is not explicitly supported, giving 8 of 40 (20%) as a sensitivity figure.

Stated counts frequently failed to reconcile with the deposits. Among the 13 studies not finally graded CONTRADICTED that reported an assay-scoped participant count with a publication source, 7 matched the number of deposited biological rows and 6 did not (3 lower, 3 higher). None of the 7 had one-sample-per-participant evidence. The candidate pool also contained four separately deposited ionisation-mode assays of one sample set (MTBLS1122, MTBLS1124, MTBLS1129, MTBLS1130), so the number of candidates overstates the number of distinct studies.

The result does not depend on unread documents or on adjudication. Three named publications could not be obtained; if all three had supplied STRICT evidence, x would be 3 and the upper bound 65. Adjudication can only confirm or reject CONTRADICTED and STRICT grades, and a rejection yields UNRESOLVED.

3.4 Author review of the AI-assisted extraction

Component A.

Q3 scope agreed in 6 of 10 studies, Q3 count in 4 of 4 where both chose the assay-scoped category, and Q4 in 10 of 10 (both "not stated" throughout, a shared null that does not test sensitivity for repeated sampling). All four Q3 disagreements (MTBLS718, 11406, 2774, 13787) ran one way: the author declined to assign a stated count to the

selected assay where the extractor had, because the documents did not say which sample set the deposit held or gave a sample rather than a participant count. The author found nothing the extractor had missed. Three of the four records were already UNRESOLVED; the fourth (MTBLS11406) would move from COUNT to UNRESOLVED, reducing the sensitivity-only STRICT + COUNT figure from 5 to 4. No disagreement bears on the STRICT count.

Component B.

Of the 12 CONTRADICTED records, 11 were confirmed by an explicit statement in the cited source (the sentence the extractor had cited), none was disputed, and one (MTBLS10972, deposit text only) was judged unresolved: the deposit describes a post-meal time course without stating that multiple specimens from the same individuals entered the assay. As a sensitivity figure, documented repeated sampling is 8 of 40 (20%) rather than 9, and the CONTRADICTED total 11 rather than 12; the frozen figures in Table 2 remain primary.

3.5 Deposits removed by quality-control filtering

In 18 of the 19 human deposits whose rows had all been removed by the quality-control rule, every sample row carried `Characteristics[Sample type] = experimental blank`; in the remaining deposit, 94 of 95 rows did. In 17 of the 19, other deposited fields identified the rows as the study's biological samples rather than blanks (V2 passed): disease-group or treatment factor values on the same rows, sample-collection protocols, or the study description. In 16 of these 17 the same fields describe participant specimens; in MTBLS15165 they describe stored platelet units whose donors are not mentioned, so the human origin of the material is not established (V1 not established). Supplementary Table S3 lists the supporting fields per deposit. In these 17 the sample-type annotation conflicts with the other fields. Three deposits were verified (all four checks passed, with stated sample counts of 368, 144 and 85), eight failed (cell lines or non-human rows in two, pooled material in two, repeated sampling in four) and eight were unverifiable, mostly because no sample count was stated.

Two further inconsistencies were found. In both assay files of MTBLS14421, the extract, assay and normalisation names of 50 of 53 rows carry the label of a different sample, and in 30 rows that label belongs to the other study group (15 control rows carry patient labels and 15 patient rows carry control labels). This is a linkage inconsistency between the sample sheet and the assay table; the deposit does not show which label is correct, and we make no claim about the published analysis. MTBLS14358 and MTBLS14376 share a title and a cohort description; whether their sample sets overlap is not established from the deposits.

3.6 Implementation limitations

The frozen implementation had three limitations, each disclosed in the result records and described in the Supplementary Note: one reading view included a differential-metabolite table whose caption escaped the exclusion pattern (nothing was cited from it); the PDF view heuristic did not recognise one heading variant (no graded record

depended on it); and the count check accepted a deposit integer when the participant-count question also cited a publication (one record, whose publication stated the component group sizes).

4 Discussion

The question is not whether the original investigators sampled acceptably, but whether a secondary analyst, working from the deposit and its linked documentation, can reconstruct the sampling structure well enough to identify independent units and account for dependence. The four analyses have different targets: the census and the documentary assessment tested a one-sample-per-participant criterion, the Workbench audit one metadata endpoint of 104 selected studies, the pre-verification sample classification rather than participant linkage. None establishes a repository-wide prevalence. Together they give convergent evidence that, under an affirmative-evidence standard, the information needed to identify independent sampling units is frequently absent from deposited metadata and often absent from linked publications. We draw three implications.

Missing subject-level linkage prevents correct dependence modelling.

A repeated-measures design is not a defect; it becomes one for reuse only when the deposit does not say which samples share a participant. In the MetaboLights census no inspected record carried a person-identifier column, and in the 104 Workbench studies examined no factor key named a subject. The documentary tranche showed what this costs: 9 of 40 randomly selected studies (22.5%) documented repeated sampling of the same participants in the deposited assay (serial sampling, pre/post samples, several specimens per patient), and a further 3 (7.5%) consisted of cell lines rather than participants. For the nine, no qualifying participant-to-sample mapping was identified under the prespecified metadata and documentary procedures. An analyst treating these rows as independent would be pseudoreplicating (Hurlbert, 1984; Waldron et al, 2016); an analyst wanting to model the dependence correctly could not. The same gap applies at the study level: four separately deposited ionisation-mode assays of one sample set (MTBLS1122, MTBLS1124, MTBLS1129, MTBLS1130) would be counted as four independent validation units by any procedure that counts accessions, and two pre-verification deposits sharing a title and cohort description (MTBLS14358, MTBLS14376) could not be told apart from independent cohorts, because nothing in either deposit states whether their sample sets overlap.

Automated metadata labels can conceal valid biological specimens.

The census classified every sample row of 220 studies as quality-control material; in 183 of them the trigger was a sample-type field reading `blank` on every row, and the 19 human deposits examined in the pre-verification were drawn from that subset. For those 19, other fields on the same rows (disease-group factors, collection protocols, a named publication) identified study samples rather than blanks in 17, and participant specimens in 16 of those. The label appears to be a template default rather than a description. A screen that trusts the label discards usable studies; one that ignores

it cannot distinguish blanks from specimens. Only the depositor can resolve this, which argues for vocabulary defaults that are empty rather than pre-filled. The linkage inconsistency in MTBLS14421, where assay rows carry the labels of other samples across study groups, is a different kind of defect: it is detectable from the deposit, but not resolvable from it.

Reproducible repository screening needs explicit evidence standards.

A screen that accepts unique sample names as evidence of independent participants would have passed hundreds of census studies; ours passed none, and the documentary tranche showed why that stricter standard was warranted, since 22.5% of the uniquely labelled candidates documented repeated sampling and 7.5% were not participant samples at all. Whatever standard a reuse study adopts should be stated before screening, applied by code that refuses other inputs, and reported with its failure categories, so that the gap between "documentation establishes independence" and "samples are independent" stays visible. The limits of such screening are also visible here: 11 of 40 randomly selected studies had no retrievable publication text, and the STRICT criterion, designed for an analysis needing one sample per participant, is stricter than the reconstructibility question; a zero STRICT count is not zero reconstructible studies.

Limitations.

The randomised documentary assessment covers one repository, and its 40 studies were drawn from the 371 unresolved candidates that the census had already filtered (human, single-matrix, at least 50 labelled sources), not from all MetaboLights studies; its design-based bound refers to that population, and its point estimate is not the result. The Workbench analysis is secondary, restricted to one metadata endpoint and to studies selected for another purpose. Evidence extraction was AI-assisted. The author's review (Section 3.4) found no case in which the extractor missed a count or a sampling statement, and its disagreements all made the frozen records more, not less, conservative; but it was performed by the author, not an independent reviewer, after partial exposure to a second model's answers, and its Q4 agreement is agreement on a shared negative, which says nothing about sensitivity for detecting repeated sampling. Component B was a consistency check by the reader who had adjudicated the same records. The review therefore supports the zero-STRICT finding and the direction of the classifications; it does not establish extraction accuracy against an independent human standard. The frozen implementations had the limitations listed in Section 3.6. Documentation can be missing for reasons unrelated to the study's design, and none of our results estimates how many deposited studies are in fact composed of independent participants.

Recommendations.

Repositories should require, at deposition, a participant identifier or a subject-to-sample mapping on every biological sample, as controlled-access genomic archives already do (National Center for Biotechnology Information, n.d.), and should flag deposits that share a sample set. Sample-type vocabularies should have no pre-filled default. Our recommendation concerns deposition metadata rather than manuscript

reporting. The STROBE-MetEpi reporting guideline for metabolomic epidemiology, currently under review (Kelly et al, 2026), already asks authors to report participant eligibility and sampling (item 5a), the biological sample collection protocol including timing and whether conditions were the same for every sample (6a), any independent validation population (17), the final number of individuals in the primary analyses (19a), the time points at which profiles were measured in cohort studies (21c), and repository access (31a). None of its 31 items asks for a machine-readable mapping from deposited samples to participants, and a manuscript statement that 100 participants contributed cannot be joined to deposited rows unless the deposit carries that mapping. A deposition-level linkage requirement would complement, not replace, this manuscript-level guidance. (The guideline's scope excludes methodological studies, so the present audit is not itself assessed against it.) None of this requires investigators to abandon repeated measurements; it requires that the deposit say where they are.

Declarations

- Funding: This research received no specific funding from any funding agency in the public, commercial or not-for-profit sectors.
- Competing interests: The author declares no competing interests.
- Ethics: the study analysed publicly deposited metadata and published documents only; no participants were recruited, no biological samples were handled, and no participant-level data were accessed or recorded. Ethical approval was not required.
- Use of AI tools: see Section 2.8.
- Supplementary Information: Supplementary Methods S1–S5 and Supplementary Tables S1–S3 (file `audit_supplement.pdf`).
- Data and code availability: protocols, code, locks, sealed evidence inventories, evidence records, grades and result records are public in the repository https://github.com/Anas-Enoch/Pathway_Identifiability (commits `b5cd507`, `3d376d0`, `195a60d`, `566af9d`; author-review records in `c27ec53`; the source of this manuscript as submitted is archived under the release tag `audit-paper-v0.7`). Full texts of publications are not redistributed; their SHA-256 values and retrieval URLs are recorded.
- Author contribution: A.E. designed the study, executed the protocols, adjudicated and reviewed the records, and wrote the manuscript.

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